higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

19 November 2019 (X-Paper)
09:00–12:00

Scientific calculators may be used.

This question paper consists of 5 pages and formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL intermediate steps and simplify where possible.
5. ALL final answers must be rounded off to THREE decimal places.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Questions must be answered in BLUE or BLACK ink.
8. Sketches must be large, neat and fully labelled.
9. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limits:

1.1.1 $\lim_{x\to\infty} \frac{x^2+5x-2}{x}$ (2)

1.1.2 $\lim_{x\to\infty} \frac{x}{x^2+1}$ (2)

1.2 Given: $-\log y = \lim_{x\to 4} \frac{2-\sqrt{x}}{4-x}$, calculate the numerical value of:

1.2.1 $\log y$ (3)

1.2.2 y (1)

1.3 Determine the value(s) of x for which $f(x)$ is discontinuous if $f(x) = \frac{1}{\cos x - 1}$; $x \in [0; 2\pi]$ (2)

[10]

QUESTION 2

2.1 Determine $\frac{dy}{dx}$ if $f(x) = \cos x$ using first principles.
Show ALL intermediate steps

HINT: $\lim_{x\to 0} \frac{\cos h - 1}{h} = 0$ and $lim_{x\to 0} \frac{\sin h}{h} = 1$ (5)

2.2 Calculate $\frac{dy}{dx}$ of the following, by making use of the derivatives of $\sin x$ and $\cos x$, as well as the rules of differentiation:

$y = 3^{\log_3 \operatorname{cosec} x}$

HINT: $a^{\log_a x} = x$ (4)

2.3 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification NOT required)

2.3.1 $y = \tan^3 \sqrt{\cot 7x}$ (4)

2.3.2 $y = 10(1 + (2 - (6 + 7x^4)^9)^3)^5$ (4)

2.4 Calculate $\frac{dy}{dx}$ if $y = \sqrt{x}^{\sqrt{x}} e^{x^2}$ with aid of logarithmic differentiation (5)

2.5 Determine $\frac{dy}{dx}$ of implicit function $10x^4 - 18xy^2 + 10y^3 = 48$ (3)

[25]

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QUESTION 3

3.1 Given: $f(x) = x^3 - x^2 - x - 1$

3.1.1 Determine the coordinates of the turning points of $f(x)$ (2)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -2$ to $x = 2$ (3)

3.1.3 Draw a neat graph of $f(x)$ between these values show the turning points on it. (2)

3.1.4 One root of the equation $x^3 - x^2 - x - 1 = 0$ is close to 1.9.

Use Taylor's/Newton's method ONCE to determine a better approximation of this root. (Correct root to THREE decimal figures) (3)

3.2 Air is being pumped into a spherical balloon in such a way that its radius increases at a rate of 0.75m/min. Find the rate of change of its volume when the radius is 5metres.

HINT: $V = \frac{4}{3}\pi r^3$ (4)

[14]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases

4.1.1 $y = [\ln^2(x^3 + 1)]\dfrac{x^2}{x^3 + 1}$ (3)

4.1.2 $y = \dfrac{\sin^{-1} 4x}{\sqrt{1 - 16x^2}}$ (3)

4.1.3 $y = \dfrac{x^3 - 4x^2 + 2x - 3}{x + 2}$ (5)

4.1.4 $y = \dfrac{1}{\text{cosec}^3 x}$ (4)

4.1.5 $y = \dfrac{\ln x}{x}$ (3)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$y = \dfrac{6x + 12}{x^2 + 5x + 6}$ (4)

[22]

QUESTION 5

5.1 Evaluate the definite integral : $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} x \sin x \, dx$ (5)

5.2 Given: $y = 4x - x^2$ and $y = x^2$

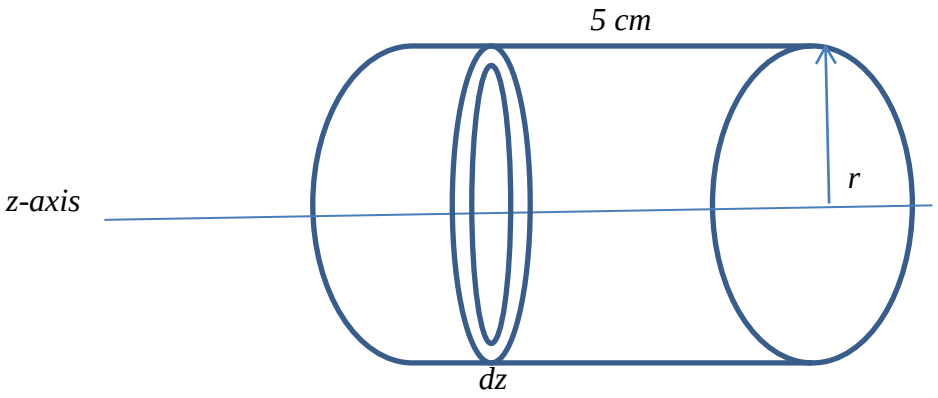
5.2.1 Calculate the coordinates of the points of intersection (2)

5.2.2 Make a neat sketch to show the enclosed area, the representative strip and the point of intersection. (2)

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2. (3)

5.2.4 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.2 is rotated about the x-axis. (4)

5.3 Calculate the moment of inertia of a flywheel with a of radius 50 cm and a thickness of 5 cm about an axis through its centre and perpendicular to the flywheel. The mass of the flywheel is 20 kg.



(4)
[20]

QUESTION 6

6.1 Determine the general solution of; $(x^2 - 1)y^3 \, dx + x^2 \, dy = 0$ (3)

6.2 Determine the particular solution of ; $\frac{1}{\sin x} . \frac{d^2y}{dx^2} = \cot x + \frac{e^{-3x}}{\sin x}$ if $\frac{dy}{dx} = \frac{5}{3}, x = 0$ and $y = \frac{1}{9}$ (6)
[9]

TOTAL : 100

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MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\cos 2x = 1 - \sin 2x$

$\sin^2 x + \cos^2 x = 1 \rightarrow \sin 2x = 1 - \cos 2x$

$1 + \tan^2 x = \sec^2 x \xrightarrow{\tan 2x} \sec 2x - 1$

$1 + \cot^2 x = \operatorname{cosec}^2 x \rightarrow \cot 2x = \operatorname{cosec} 2x - 1$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = \frac{1}{2} - \frac{1}{2} \cos 2A = \frac{1}{2}(1 - \cos 2A)$

$\cos^2 A = \frac{1}{2} + \frac{1}{2} \cos 2A = \frac{1}{2}(1 + \cos 2A)$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin A \cos B = \frac{1}{2} [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = \frac{1}{2} [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = \frac{1}{2} [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = \frac{1}{2} [\cos (A - B) - \cos (A + B)]$

$$\tan x = \frac{\sin x}{\cos x}\ ;\ \sin x = \frac{1}{\operatorname{cosec} x}\ ;\ \cos x = \frac{1}{\sec x}$$

$$\cot x = \frac{\cos x}{\sin x}\ ;\ \operatorname{cosec} x = \frac{1}{\sin x}\ ;\ \sec x = \frac{1}{\cos x}$$

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BINOMIAL THEOREM

$$(x+h)^n = x^n + nx^{n-1}\,h + \frac{n(n-1)}{2!}x^{n-2}\,h^2 + \frac{n(n-1)(n-2)}{1\times 2\times 3}x^{n-3}h^3 + \ldots\ldots$$

DI**FFERENTIATION**

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

y = u(x) . v(x)

$$\frac{dy}{dx} = u\,.\frac{dv}{dx} + v\,.\frac{du}{dx}$$
$$= u\cdot v' + v\cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\,.\frac{du}{dx} - u\,.\frac{dv}{dx}}{v^2}$$

$$= \frac{v\cdot u' - u\cdot v'}{v^2}$$

CHAIN RULE

y = f(u(x))

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

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f(x)	$\frac{d}{dx} f(x)$	$\int f(x)dx$
xn	nx^{n-1}	$\frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$
axn	$a\frac{d}{dx} x^n$	$a\int x^n dx$
eax+b	eax+b . $\frac{d}{dx}(ax+b)$	$\frac{e^{ax+b}}{\frac{d}{dx}(ax+b)} + C$
adx+e	adx+e.lna $\frac{d}{dx}(dx+e)$	$\frac{a^{dx+e}}{\ln a.\frac{d}{dx}(dx+e)} + C$
ln(ax)	$\frac{1}{ax}.\frac{d}{dx} ax$	xln ax – x + C
$e^{f(x)}$	$e^{f(x)} . \frac{d}{dx} f(x)$	—
$a^{f(x)}$	$a^{f(x)} .\ln a. \frac{d}{dx} f(x)$	—
$ln\ f(x)$	$\frac{1}{f(x)}.\frac{d}{dx} f(x)$	—
sin ax	a cos ax	$-\frac{\cos ax}{a}. + C$
cos ax	-a sin ax	$\frac{\sin ax}{a}. + C$
tan ax	a sec2 ax	$\frac{1}{a}\ln[\sec(ax)] + C$
cot ax	- a cosec2 ax	$\frac{1}{a}\ln[\sin(ax)] + C$
sec ax	a sec ax·tan ax	$\frac{1}{a}$ ln [sec ax + tan ax] + C
cosec ax	- a cosec ax·cot ax	$\frac{1}{a}\ln\left[\tan\left(\frac{ax}{2}\right)\right] + C$

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f(x)	$\frac{d}{dx} f(x)$	$\int f(x)dx$
sin f(x)	cos f(x)•f 1(x)	___
cos f(x)	- sin f(x)•f 1(x)	___
tan f(x)	sec2 f(x)•f 1(x)	___
cot f(x)	- cosec2 f(x)•f 1 (x)	___
sec f(x)	sec f(x)•tan f(x)•f 1(x)	___
cosec f(x)	- cosec f(x)•cot f(x)•f 1(x)	___
$\sin^{-1} f(x)$	$\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$	__
$\cos^{-1} f(x)$	$\frac{-f'(x)}{\sqrt{1-[f(x)]^2}}$	__
$\tan^{-1} f(x)$	$\frac{f'(x)}{1+[f(x)]^2}$	__
$\cot^{-1} f(x)$	$\frac{-f'(x)}{1+[f(x)]^2}$	__
$\sec^{-1} f(x)$	$\frac{f'(x)}{f(x)\sqrt{[f(x)]^2-1}}$	
$\text{cosec}^{-1} f(x)$	$\frac{-f'(x)}{f(x)\sqrt{[f(x)]^2-1}}$	__
sin2(ax)	____	$\frac{x}{2}-\frac{\sin(2ax)}{4a}+C$
cos2(ax)	____	$\frac{x}{2}+\frac{\sin(2ax)}{4a}+C$
tan2(ax)	____	$\frac{1}{a}\tan(ax)-x+C$
cot2(ax)	_	$-\frac{1}{a}\cot(ax)-x+C$

INTEGRATION

$\int f(x)\, g'(x)\, dx =$ f(x) g(x) - $\int$ f '(x) g(x) dx

$\int [f(x)]^n \bullet f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + C$

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APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b y\,dx \; ; \; A_x = \int_a^b (y_1 - y_2)\,dx$$

$$A_y = \int_a^b x\,dy \; ; \; A_y = \int_a^b (x_1 - x_2)\,dy$$

VOLUMES

$$V_x = \pi \int_a^b y^2 dx \; ; V_x = \pi \int_a^b \left(y_1^2 - y_2^2\right) dx$$

$$V_y = \pi \int_a^b x^2 dy \; ; V_y = \pi \int_a^b \left(x_1^2 - x_2^2\right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 dA \; ; \; I_y = \int_a^b r^2 dA$$

MOMENTS OF INERTIA

Mass = density × volume

M = ρV

DEFINITION: I = m r2

GENERAL: $I = \int_a^b r^2 dm = \rho \int_a^b r^2 dV$

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